

Parcial 2 – Redes

00103923 – Gabriel Alejandro Batres Flores

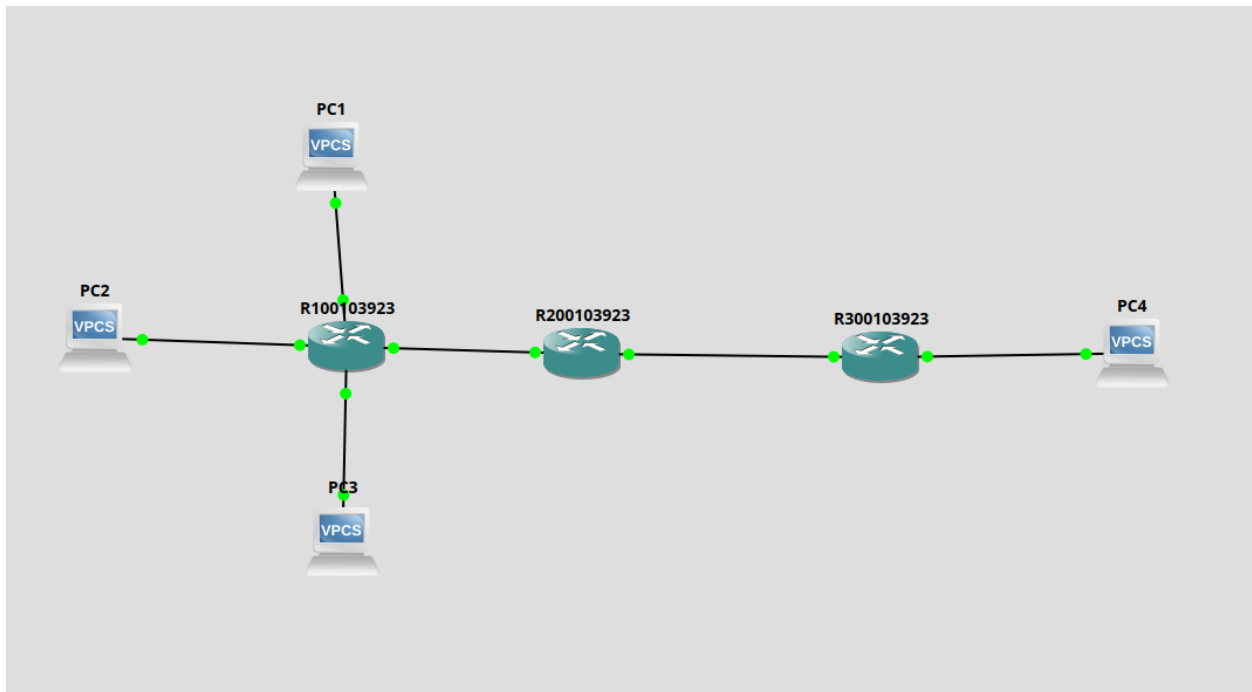
Ejercicio 1

Carnet: 00103923

A: 32

B: 51

Red base asignada: 10.32.51.0/24



Interface fastEthernet

R1:

```

R100103923#enable
R100103923#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R100103923(config)#hostname R1
R1(config)#no ip domain-lookup
R1(config)#interface FastEthernet0/0
R1(config-if)#ip address 10.32.51.1 255.255.255.192
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#inter
*Mar 1 00:02:44.235: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state t
o up
*Mar 1 00:02:45.235: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/0, changed state to up
R1(config)#interface FastEthernet0/1
R1(config-if)#ip address 10.32.51.65 255.255.255.224
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#inte
*Mar 1 00:03:25.787: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state t
o up
*Mar 1 00:03:26.787: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/1, changed state to up
R1(config)#interface FastEthernet1/0
R1(config-if)#ip address 10.32.51.97 255.255.255.240
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#inte
*Mar 1 00:04:11.467: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state t
o up
*Mar 1 00:04:12.467: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et1/0, changed state to up
R1(config)#interface FastEthernet2/0
R1(config-if)#ip address 10.32.51.121 255.255.255.252
% Invalid input detected at '^' marker.

R1(config-if)#ip address 10.32.51.121 255.255.255.252
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
*Mar 1 00:05:44.379: %LINK-3-UPDOWN: Interface FastEthernet2/0, changed state t
o up
*Mar 1 00:05:45.379: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et2/0, changed state to up
R1(config)#

```

R2:

```

R200103923#enable
R200103923#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R200103923(config)#hostname R2
R2(config)#no ip domain-lookup
R2(config)#interface fastEthernet0/0
R2(config-if)#ip address 10.32.51.122 255.255.255.252
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#inter
*Mar 1 00:05:21.927: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state t
o up
*Mar 1 00:05:22.927: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/0, changed state to up
R2(config)#interface fastEthernet0/1
R2(config-if)#ip address 10.32.51.125 255.255.255.252
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#
*Mar 1 00:05:58.707: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar 1 00:05:59.707: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
R2(config)#

```

R3:

```

R300103923#enable
R300103923#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R300103923(config)#hostname R3
R3(config)#no ip domain-lookup
R3(config)#interface fastEthernet
% Incomplete command.

R3(config)#interface fastEthernet0/0
R3(config-if)#ip address 10.32.51.113 255.255.255.248
R3(config-if)#no shutdown
R3(config-if)#exit
*Mar 1 00:07:38.907: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state t
o up
*Mar 1 00:07:39.907: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/0, changed state to up
R3(config-if)#exit
R3(config)#interface fastEthernet0/1
R3(config-if)#ip address 10.32.51.126 255.255.255.252
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#
*Mar 1 00:08:35.831: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state t
o up
*Mar 1 00:08:36.831: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/1, changed state to up
R3(config)#

```

Ip PCs:

PC1:

```

PC1> ip 10.32.51.2 255.255.255.192 10.32.51.1
Checking for duplicate address...
PC1 : 10.32.51.2 255.255.255.192 gateway 10.32.51.1

PC1> save
Saving startup configuration to startup.vpc
. done

PC1>

```

PC2:

```

PC2> ip 10.32.51.66 255.255.255.224 10.32.51.65
Checking for duplicate address...
PC2 : 10.32.51.66 255.255.255.224 gateway 10.32.51.65

PC2> save
Saving startup configuration to startup.vpc
. done

PC2>

```

PC3:

```

PC3> ip 10.32.51.98 255.255.255.240 10.32.51.97
Checking for duplicate address...
PC3 : 10.32.51.98 255.255.255.240 gateway 10.32.51.97
PC3> save
Saving startup configuration to startup.vpc
. done
PC3>

```

PC4:

```

PC4> ip 10.32.51.114 255.255.255.248 10.32.51.113
Checking for duplicate address...
PC4 : 10.32.51.114 255.255.255.248 gateway 10.32.51.113
PC4> save
Saving startup configuration to startup.vpc
. done
PC4>

```

Enrutamiento estático:

R1:

```

R1(config)#ip route 10.32.51.112 255.255.255.248 10.32.51.122
R1(config)#exit
R1#
*Mar  1 00:19:47.747: %SYS-5-CONFIG_I: Configured from console by console
R1#

```

R2:

```

R2(config)#ip route 10.32.51.0 255.255.255.192 10.32.51.121
R2(config)#ip route 10.32.51.64 255.255.255.224 10.32.51.121
R2(config)#ip route 10.32.51.96 255.255.255.240 10.32.51.121
R2(config)#ip route 10.32.51.112 255.255.255.248 10.32.51.126
R2(config)#exit
R2#
*Mar  1 00:20:33.675: %SYS-5-CONFIG_I: Configured from console by console
R2#

```

R3:

```

R3#
*Mar  1 00:08:36.831: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
R3(config)#ip route 10.32.51.0 255.255.255.192 10.32.51.125
R3(config)#ip route 10.32.51.64 255.255.255.224 10.32.51.125
R3(config)#ip route 10.32.51.96 255.255.255.240 10.32.51.125
R3(config)#exit
R3#

```

Prueba:

```
PC4
Trying ::1...
Connected to localhost.
Escape character is '^]'.

PC4> ping 10.32.51.2

84 bytes from 10.32.51.2 icmp_seq=1 ttl=61 time=47.854 ms
84 bytes from 10.32.51.2 icmp_seq=2 ttl=61 time=37.602 ms
84 bytes from 10.32.51.2 icmp_seq=3 ttl=61 time=36.580 ms
84 bytes from 10.32.51.2 icmp_seq=4 ttl=61 time=36.877 ms
84 bytes from 10.32.51.2 icmp_seq=5 ttl=61 time=36.605 ms

PC4> trace 10.32.51.2
trace to 10.32.51.2, 8 hops max, press Ctrl+C to stop
 1  10.32.51.113    5.343 ms  10.101 ms  9.446 ms
 2  10.32.51.125   19.993 ms 19.822 ms 19.768 ms
 3  10.32.51.121   29.775 ms 30.058 ms 29.278 ms
 4  *10.32.51.2    39.791 ms (ICMP type:3, code:3, Destination port unreachable)

PC4> █
```

...
*Mar 1 00:19:47.747: %SYS-5-CONFIG_I: Configured from console by console

R1#show ip interface brief

Interface	IP-Address	OK?	Method	Status	Prot
ocol					
FastEthernet0/0	10.32.51.1	YES	manual	up	up
FastEthernet0/1	10.32.51.65	YES	manual	up	up
FastEthernet1/0	10.32.51.97	YES	manual	up	up
FastEthernet2/0	10.32.51.121	YES	manual	up	up

R1#show ip route

Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

```
10.0.0.0/8 is variably subnetted, 5 subnets, 5 masks
C    10.32.51.0/26 is directly connected, FastEthernet0/0
C    10.32.51.64/27 is directly connected, FastEthernet0/1
C    10.32.51.120/30 is directly connected, FastEthernet2/0
S    10.32.51.112/29 [1/0] via 10.32.51.122
C    10.32.51.96/28 is directly connected, FastEthernet1/0
R1#█
```

```

R2#show ip interface brief
Interface          IP-Address      OK? Method Status      Prot
ocol
FastEthernet0/0    10.32.51.122    YES manual up          up
FastEthernet0/1    10.32.51.125    YES manual up          up
FastEthernet1/0    unassigned      YES unset  administratively down down
FastEthernet2/0    unassigned      YES unset  administratively down down

```

```

R2#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

```

Gateway of last resort is not set

```

      10.0.0.0/8 is variably subnetted, 6 subnets, 5 masks
S       10.32.51.0/26 [1/0] via 10.32.51.121
S       10.32.51.64/27 [1/0] via 10.32.51.121
C       10.32.51.120/30 is directly connected, FastEthernet0/0
C       10.32.51.124/30 is directly connected, FastEthernet0/1
S       10.32.51.112/29 [1/0] via 10.32.51.126
S       10.32.51.96/28 [1/0] via 10.32.51.121

```

R2#

```

S       10.32.51.96/28 [1/0] via 10.32.51.125
R3#show ip interface brief
Interface          IP-Address      OK? Method Status      Prot
ocol
FastEthernet0/0    10.32.51.113    YES manual up          up
FastEthernet0/1    10.32.51.126    YES manual up          up
FastEthernet1/0    unassigned      YES unset  administratively down down
FastEthernet2/0    unassigned      YES unset  administratively down down

```

```

R3#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

```

Gateway of last resort is not set

```

      10.0.0.0/8 is variably subnetted, 5 subnets, 5 masks
S       10.32.51.0/26 [1/0] via 10.32.51.125
S       10.32.51.64/27 [1/0] via 10.32.51.125
C       10.32.51.124/30 is directly connected, FastEthernet0/1
C       10.32.51.112/29 is directly connected, FastEthernet0/0
S       10.32.51.96/28 [1/0] via 10.32.51.125

```

R3#

Calculos VSLM:

Ejercicio 1.					
Carnet 00103923					
A=32					
B=51					
Red base asignada: 10.32.51.0/24					
PC1 PC2 - R1 - R2 - R3 - PC4 PC3					
PC1	50 → 2 ⁶ → 64	prefijo: 32-6 = /26	255.255.255.192/26	S=64	
PC2	25 → 2 ⁵ → 32	" : 32-5 = /27	255.255.255.224/27	S=32	
PC3	12 → 2 ⁴ → 16	" : 32-4 = /28	255.255.255.240/28	S=16	
PC4	6 → 2 ³ → 8	" : 32-3 = /29	... 248/29	S=8	
Red	Primera	Ultima	Broadcast	Mascara	
PC1	10.32.51.0	10.32.51.1	10.32.51.63	255.255.255.192	
PC2	10.32.51.64	10.32.51.65	10.32.51.95	224	
PC3	96	97	110	111	240
PC4	112	113	118	119	248
E1-2	120	121	122	123	252
E2-3	124	125	126	127	252
R1: 0/0 → PC1; 0/1 → PC2; 1/0 → PC3; 2/0 → R2.					
R2: 0/0 → R1; 0/1 → R3					
R3: 0/0 → PC4; 0/1 → R2.					

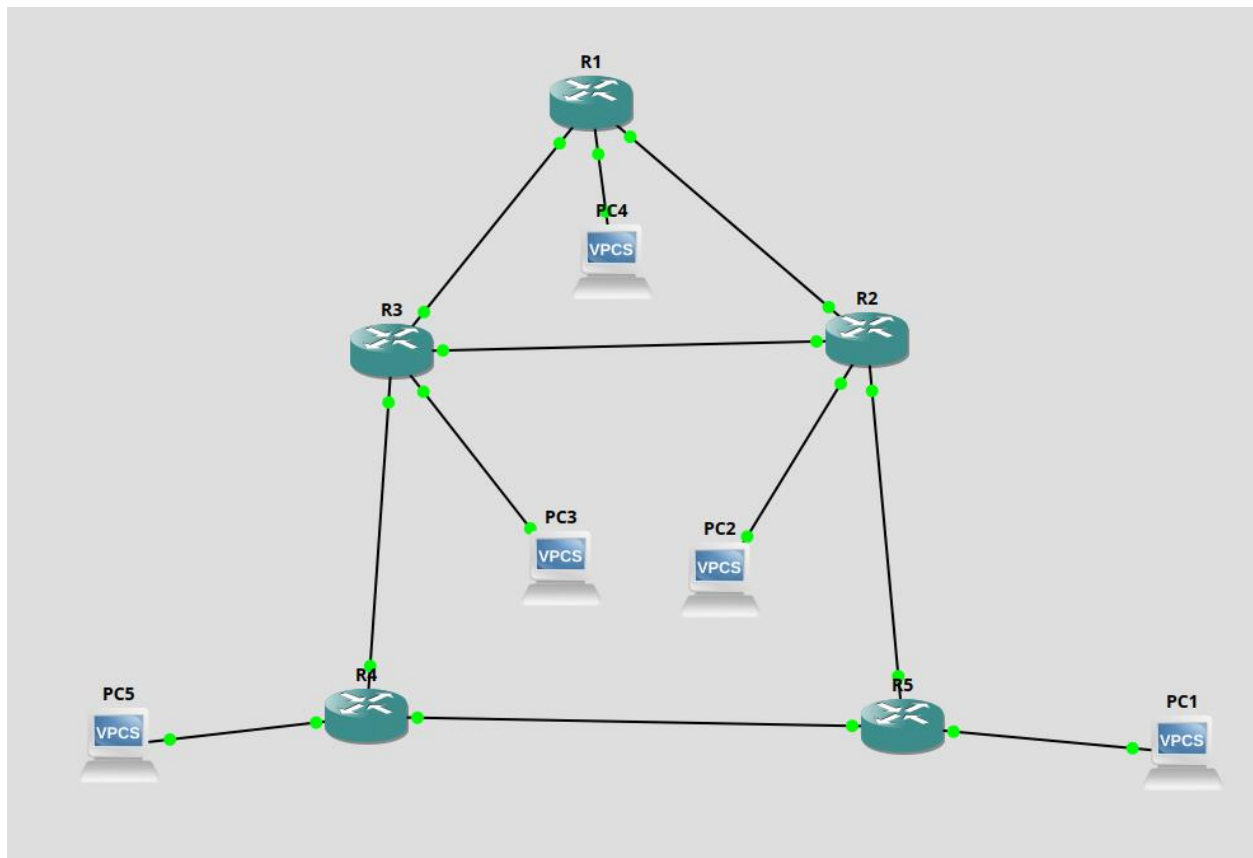
Ejercicio 2

Carnet: 00103923

A: 32

B: 51

Red base asignada: 192.32.51.0/24



Interface fastEthernet

R1:


```

R1#enable
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#hostname R100103923
R100103923(config)#no ip domain -lookup
^
% Invalid input detected at '^' marker.

R100103923(config)#no ip domain-lookup
R100103923(config)#interface fastEthernet0/0
^
% Invalid input detected at '^' marker.

R100103923(config)#interface fastEthernet0/0
R100103923(config-if)#ip address 192.32.51.113 255.255.255.240
R100103923(config-if)#no shutdown
R100103923(config-if)#exit
R100103923(config)#
*Mar  1 00:04:18.215: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state t
o up
*Mar  1 00:04:19.215: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/0, changed state to up
R100103923(config)#interface fastEthernet0/1
R100103923(config-if)#ip address 192.32.51.137 255.255.255.252
R100103923(config-if)#no shutdown
R100103923(config-if)#exit
R100103923(config)#int
*Mar  1 00:05:12.895: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state t
o up
*Mar  1 00:05:13.895: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/1, changed state to up
R100103923(config)#interface fastEthernet1/0
R100103923(config-if)#ip address 192.32.51.141 255.255.255.252
R100103923(config-if)#no shutdown
R100103923(config-if)#exit
R100103923(config)#
*Mar  1 00:06:05.319: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state t
o up
*Mar  1 00:06:06.319: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et1/0, changed state to up
R100103923(config)#

```

R2:

00:07:48:279: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to down.

R2#enable

R2#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2(config)#interface fastEthernet0/0

R2(config-if)#192.32.51.65 255.255.255.224

^
% Invalid input detected at '^' marker.

R2(config-if)#ip address 192.32.51.65 255.255.255.224

R2(config-if)#no shutdown

R2(config-if)#exit

R2(config)#

*Mar 1 00:05:48.279: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to
up

*Mar 1 00:05:49.279: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

R2(config)#interface fastEthernet0/1

R2(config-if)#ip address 192.32.51.157 255.255.255.252

R2(config-if)#no shutdown

R2(config-if)#exit

R2(config)#interface

*Mar 1 00:06:55.871: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to
up

*Mar 1 00:06:56.871: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

R2(config)#interface fastEthernet1/0

R2(config-if)#ip address 192.32.51.142 255.255.255.252

R2(config-if)#no shutdown

R2(config-if)#exit

R2(config)#interfac

*Mar 1 00:07:28.847: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to
up

*Mar 1 00:07:29.847: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up

R2(config)#interface fastEthernet2/0

R2(config-if)#192.32.51.145 255.255.255.252

^
% Invalid input detected at '^' marker.

R2(config-if)#ip address 192.32.51.145 255.255.255.252

R2(config-if)#no shutdown

R2(config-if)#exit

R2(config)#

*Mar 1 00:08:09.911: %LINK-3-UPDOWN: Interface FastEthernet2/0, changed state to
up

*Mar 1 00:08:10.911: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/0, changed state to up

R2(config)#

R3:

```

Mar 1 00:00:05.943: %LINK-5-CHANGED: Interface FastEthernet2/0, changed state
to administratively down
*Mar 1 00:00:05.943: %LINK-5-CHANGED: Interface FastEthernet2/0, changed state
to administratively down
R3#enable
R3#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#hostname 300103923
% Hostname contains one or more illegal characters.

300103923(config)#no ip domain-lookup
300103923(config)#hostname R300103923
R300103923(config)#interface fastEthernet0/0
R300103923(config-if)#ip address 192.32.51.97 255.255.255.240
R300103923(config-if)#no shutdown
R300103923(config-if)#exit
R300103923(config)#inte
*Mar 1 00:08:31.843: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state t
o up
*Mar 1 00:08:32.843: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/0, changed state to up
R300103923(config)#interface fastEthernet0/1
R300103923(config-if)#ip address 192.32.51.138 255.255.255.252
R300103923(config-if)#no shutdown
R300103923(config-if)#exit
R300103923(config)#interface
*Mar 1 00:09:16.199: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state t
o up
*Mar 1 00:09:17.199: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/1, changed state to up
R300103923(config)#interface fastEthernet1/0
R300103923(config-if)#ip address 192.32.51.149 255.255.255.252
R300103923(config-if)#no shutdown
R300103923(config-if)#exit
R300103923(config)#int
*Mar 1 00:09:46.259: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state t
o up
*Mar 1 00:09:47.259: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et1/0, changed state to up
R300103923(config)#interface fastEthernet2/0
R300103923(config-if)#ip address 192.32.51.146 255.255.255.252
R300103923(config-if)#no shutdown
R300103923(config-if)#exit
R300103923(config)#
*Mar 1 00:10:23.143: %LINK-3-UPDOWN: Interface FastEthernet2/0, changed state t
o up
*Mar 1 00:10:24.143: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et2/0, changed state to up
R300103923(config)#

```

R4:

```

*Mar  1 00:00:03.333: %LINK-5-CHANGED: Interface FastEthernet0/0, changed state
to administratively down
*Mar  1 00:00:03.375: %LINK-5-CHANGED: Interface FastEthernet1/0, changed state
to administratively down
*Mar  1 00:00:03.383: %LINK-5-CHANGED: Interface FastEthernet2/0, changed state
to administratively down
*Mar  1 00:00:04.307: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/0, changed state to down
*Mar  1 00:00:04.339: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/1, changed state to down
*Mar  1 00:00:04.375: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et1/0, changed state to down
*Mar  1 00:00:04.383: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et2/0, changed state to down
R4#enable
R4#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R4(config)#hostname R400103923
R400103923(config)#no ip domain-lookup
R400103923(config)#interface fastEthernet0/0
R400103923(config-if)#ip address 192.32.51.128 255.255.255.248
Bad mask /29 for address 192.32.51.128
R400103923(config-if)#ip address 192.32.51.129 255.255.255.248
R400103923(config-if)#no shutdown
R400103923(config-if)#exit
*Mar  1 00:02:12.739: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state t
o up
*Mar  1 00:02:13.739: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/0, changed state to up
R400103923(config-if)#exit
R400103923(config)#interface fastEthernet0/1
R400103923(config-if)#ip address 192.32.51.153 255.255.255.252
R400103923(config-if)#no shutdown
R400103923(config-if)#exit
R400103923(config)#
*Mar  1 00:03:19.199: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state t
o up
*Mar  1 00:03:20.199: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/1, changed state to up
R400103923(config)#interface fastEthernet1/0
R400103923(config-if)#ip address 192.32.51.150 255.255.255.252
R400103923(config-if)#no shutdown
R400103923(config-if)#exit
R400103923(config)#
*Mar  1 00:03:57.299: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state t
o up
*Mar  1 00:03:58.299: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et1/0, changed state to up
R400103923(config)#

```

R5:

```

*Mar 1 00:00:04.595: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to administratively down
*Mar 1 00:00:04.595: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to down
*Mar 1 00:00:04.631: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to down
*Mar 1 00:00:04.667: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
*Mar 1 00:00:04.671: %LINK-3-UPDOWN: Interface FastEthernet2/0, changed state to down
R5#enable
R5#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R5(config)#hostname R500103923
R500103923(config)#no ip domain-lookup
R500103923(config)#interface fastEthernet0/0
R500103923(config-if)#ip address 192.32.51.1 255.255.255.192
R500103923(config-if)#no shutdown
R500103923(config-if)#exit
R500103923(config)#interface
*Mar 1 00:14:56.123: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:14:57.123: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R500103923(config)#interface fastEthernet0/1
R500103923(config-if)#ip address 192.32.51.157 255.255.255.252
R500103923(config-if)#no shutdown
R500103923(config-if)#exit
R500103923(config)#
*Mar 1 00:15:33.343: %IP-4-DUPADDR: Duplicate address 192.32.51.157 on FastEthernet0/1, sourced by c202.2aff.0001
R500103923(config)#
*Mar 1 00:15:35.319: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar 1 00:15:36.319: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
R500103923(config)#interface fastEthernet1/0
R500103923(config-if)#ip address 192.32.51.
*Mar 1 00:16:03.475: %IP-4-DUPADDR: Duplicate address 192.32.51.157 on FastEthernet0/1, sourced by c202.2aff.0001
R500103923(config-if)#ip address 192.32.51.154 255.255.255.252
R500103923(config-if)#no shutdown
R500103923(config-if)#exit
R500103923(config)#
*Mar 1 00:16:20.207: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up
*Mar 1 00:16:21.207: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up
R500103923(config)#

```

```

R500103923#show ip interface
*Mar  1 00:34:55.167: %IP-4-DUPADDR: Duplicate address 192.32.51.157 on FastEthe
rnet0/1, sourced by c202.2aff.0001
R500103923#show ip interface brief
Interface                IP-Address      OK? Method Status      Prot
ocol
FastEthernet0/0          192.32.51.1     YES manual up          up
FastEthernet0/1          192.32.51.157   YES manual up          up
FastEthernet1/0          192.32.51.154   YES manual up          up
FastEthernet2/0          unassigned       YES unset  administratively down down

R500103923#enable
R500103923#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R500103923(config)#i
*Mar  1 00:35:26.531: %IP-4-DUPADDR: Duplicate address 192.32.51.157 on FastEthe
rnet0/1, sourced by c202.2aff.0001
R500103923(config)#interface fastEthernet0/1
R500103923(config-if)#no ip address 192.32.51.157 255.255.255.252
R500103923(config-if)#ip address 192.32.51.158 255.255.255.252
R500103923(config-if)#no shutdown
R500103923(config-if)#exit
R500103923(config)#exit
R500103923#show ip interface brief
*Mar  1 00:36:25.147: %SYS-5-CONFIG_I: Configured from console by console
R500103923#show ip interface brief
Interface                IP-Address      OK? Method Status      Prot
ocol
FastEthernet0/0          192.32.51.1     YES manual up          up
FastEthernet0/1          192.32.51.158   YES manual up          up
FastEthernet1/0          192.32.51.154   YES manual up          up
FastEthernet2/0          unassigned       YES unset  administratively down down

R500103923#

```

IP PCs

PC1:

```
PC1

Welcome to Virtual PC Simulator, version 0.8.3
Dedicated to Daling.
Build time: Sep  9 2023 11:15:00
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC1> ip 192.32.51.2 255.255.255.192 192.32.51.1
Checking for duplicate address...
PC1 : 192.32.51.2 255.255.255.192 gateway 192.32.51.1

PC1> save
Saving startup configuration to startup.vpc
. done

PC1> █
```

PC2:

```
PC2

Trying ::1...
Connected to localhost.
Escape character is '^]'.

PC2> show ip

NAME       : PC2[1]
IP/MASK     : 192.32.51.66/27
GATEWAY     : 192.32.51.65
DNS         :
MAC         : 00:50:79:66:68:01
LPORT      : 10002
RHOST:PORT  : 127.0.0.1:10003
MTU         : 1500

PC2> █
```

PC3:

```
PC3

Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC3> ip 192.32.51.98 255.255.255.240 193.32.51.97
not same subnet

PC3> ip 192.32.51.98 255.255.255.240 192.32.51.97
Checking for duplicate address...
PC3 : 192.32.51.98 255.255.255.240 gateway 192.32.51.97

PC3> save
Saving startup configuration to startup.vpc
. done

PC3> █
```

PC4:

```
PC4

Welcome to Virtual PC Simulator, version 0.8.3
Dedicated to Daling.
Build time: Sep  9 2023 11:15:00
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC4> ip 192.32.51.114 255.255.255.240 192.32.51.113
Checking for duplicate address...
PC4 : 192.32.51.114 255.255.255.240 gateway 192.32.51.113

PC4> save
Saving startup configuration to startup.vpc
. done

PC4> █
```

PC5:


```
PC5
Welcome to Virtual PC Simulator, version 0.8.3
Dedicated to Daling.
Build time: Sep  9 2023 11:15:00
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC5> ip 192.32.51.130 255.255.255.248 192.32.51.129
Checking for duplicate address...
PC5 : 192.32.51.130 255.255.255.248 gateway 192.32.51.129

PC5> save
Saving startup configuration to startup.vpc
. done

PC5> █
```

OSPF

R1:

```
*****
*Mar  1 00:06:05.319: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state t
o up
*Mar  1 00:06:06.319: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et1/0, changed state to up
R100103923(config)#router ospf 1
R100103923(config-router)#network 192.32.51.112 0.0.0.15 area 0
R100103923(config-router)#network 192.32.51.140 0.0.0.3 area
% Incomplete command.

R100103923(config-router)#network 192.32.51.140 0.0.0.3 area 0
R100103923(config-router)#network 192.32.51.136 0.0.0.3 area 0
R100103923(config-router)#exit
R100103923(config)#█
```

R2:

```

FastEthernet0/0 192.32.51.149 100 manual up
R2#enable
R2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#router ospf 1
R2(config-router)#network 192.32.51.64 0.0.0.31 area 0
R2(config-router)#network 192.32.51.140 0.0.0.3 area 0
R2(config-router)#network 192.32.51. 0.0.0.3 area 0
*Mar 1 00:46:35.643: %OSPF-5-ADJCHG: Process 1, Nbr 192.32.51.141 on FastEthernet1/0 from LOADING to FULL, Loading Done
R2(config-router)#network 192.32.51.144 0.0.0.3 area 0
R2(config-router)#network 192.32.51.156 0.0.0.3 area 0
R2(config-router)#exit
R2(config)#exit
R2#sh
*Mar 1 00:47:35.863: %SYS-5-CONFIG_I: Configured from console by console
R2#

```

R3:

```

R3#sh
*Mar 1 00:10:24.143: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R300103923(config)#
*Mar 1 00:15:00.319: %IP-4-DUPADDR: Duplicate address 192.32.51.149 on FastEthernet1/0, sourced by c204.2b65.0010
R300103923(config)#router ospf 1
R300103923(config-router)#network 192.32.51.96 0.0.0.15 area 0
R300103923(config-router)#network 192.32.51.136 0.0.0.3 area 0
R300103923(config-router)#network 192.32.51.1 0.0.0.3 area 0
*Mar 1 00:47:31.467: %OSPF-5-ADJCHG: Process 1, Nbr 192.32.51.141 on FastEthernet0/1 from LOADING to FULL, Loading Done
R300103923(config-router)#network 192.32.51.140 0.0.0.3 area 0
R300103923(config-router)#network 192.32.51.148 0.0.0.3 area 0
R300103923(config-router)#exit
R300103923(config)#

```

R4:

```

R400103923(config-if)#no shutdown
R400103923(config-if)#exit
R400103923(config)#
*Mar 1 00:03:57.299: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up
*Mar 1 00:03:58.299: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up
R400103923(config)#router ospf 1
R400103923(config-router)#network 192.32.51.128 0.0.0.7 area 0
R400103923(config-router)#network 192.32.51.148 0.0.0.3 area 0
R400103923(config-router)#network 192.32.51.152 0.0.0.3 area 0
R400103923(config-router)#
*Mar 1 00:33:31.219: %OSPF-5-ADJCHG: Process 1, Nbr 192.32.51.149 on FastEthernet1/0 from LOADING to FULL, Loading Done
R400103923(config-router)#exit
R400103923(config)#

```

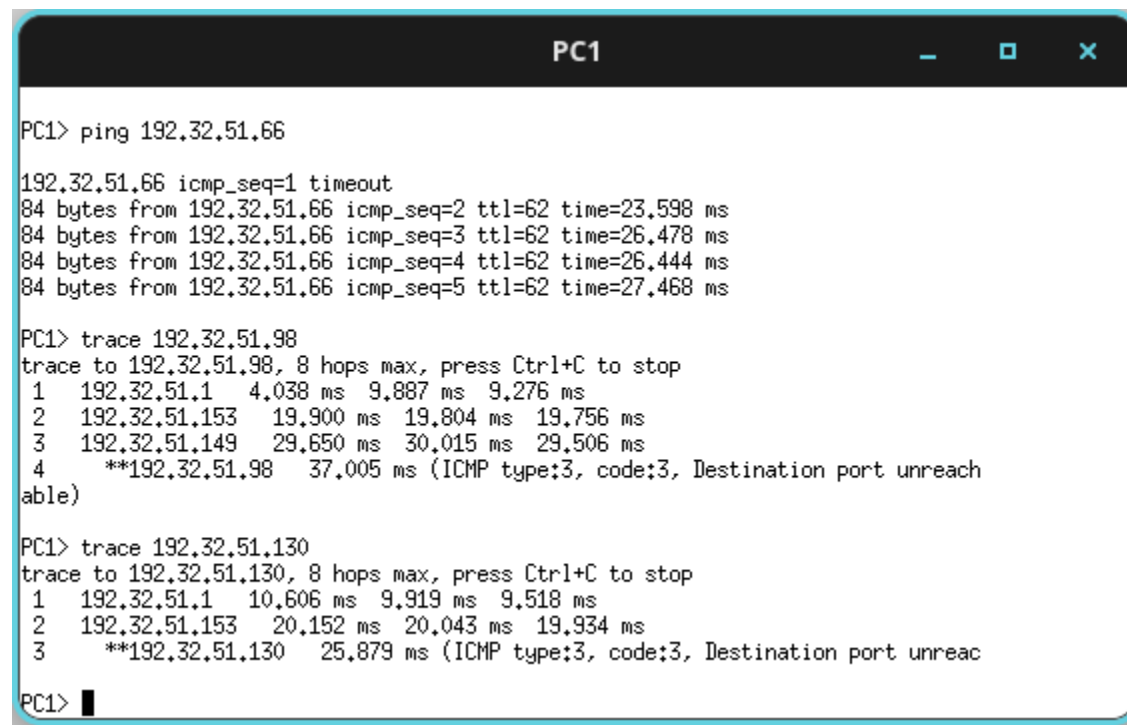
R5:

```

FastEthernet0/27/0      unassigned      123 unset      administratively down down
R500103923#enable
R500103923#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R500103923(config)#router ospf 1
R500103923(config-router)#network 192.32.51.0 0.0.0.63 area 0
R500103923(config-router)#network 192.32.51.152 0.0.0.3 area 0
R500103923(config-router)#network 192.32.51.1 0.0.0.3 area 0
*Mar 1 00:45:01.527: %OSPF-5-ADJCHG: Process 1, Nbr 192.32.51.153 on FastEthernet0/1 from LOADING to FULL, Loading Done
R500103923(config-router)#network 192.32.51.156 0.0.0.3 area 0
R500103923(config-router)#exit
R500103923(config)#
*Mar 1 00:45:08.707: %OSPF-5-ADJCHG: Process 1, Nbr 192.32.51.157 on FastEthernet0/1 from LOADING to FULL, Loading Done
R500103923(config)#

```

Pruebas



```

PC1> ping 192.32.51.66

192.32.51.66 icmp_seq=1 timeout
84 bytes from 192.32.51.66 icmp_seq=2 ttl=62 time=23.598 ms
84 bytes from 192.32.51.66 icmp_seq=3 ttl=62 time=26.478 ms
84 bytes from 192.32.51.66 icmp_seq=4 ttl=62 time=26.444 ms
84 bytes from 192.32.51.66 icmp_seq=5 ttl=62 time=27.468 ms

PC1> trace 192.32.51.98
trace to 192.32.51.98, 8 hops max, press Ctrl+C to stop
 1  192.32.51.1    4.038 ms  9.887 ms  9.276 ms
 2  192.32.51.153  19.900 ms  19.804 ms  19.756 ms
 3  192.32.51.149  29.650 ms  30.015 ms  29.506 ms
 4      **192.32.51.98    37.005 ms (ICMP type:3, code:3, Destination port unreachable)

PC1> trace 192.32.51.130
trace to 192.32.51.130, 8 hops max, press Ctrl+C to stop
 1  192.32.51.1    10.606 ms  9.919 ms  9.518 ms
 2  192.32.51.153  20.152 ms  20.043 ms  19.934 ms
 3      **192.32.51.130   25.879 ms (ICMP type:3, code:3, Destination port unreachable)

PC1>

```

Calculos VSLM

Ejercicio 2

Connet: 00103923

A: 32

B: 51

Red asignada: 192.32.51.0/24

PC1	50 → 2 ⁶ → 64	Prefijo: 32-6 = 126	255.255.255.192	0.0.0.63
PC2	25 → 2 ⁵ → 32	" 32-5 = 127	... 224	0.0.0.31
PC3	12 → 2 ⁴ → 16	" 32-4 = 128	... 240	0.0.0.15
PC4	8 → 2 ³ → 8	" 32-4 = 128	... 240	0.0.0.15
PC6	6 → 2 ³ → 8	" 32-3 = 129	... 248	0.0.0.7

PC1	192.32.51.0	192.32.51.1	192.32.51.62	192.32.51.63	255.255.255.192
PC2	.64	.65	.94	.95	.224
PC3	.96	.97	.110	.111	.240
PC4	.112	.113	.126	.127	.240
PC5	.128	.129	.134	.135	.248
E1-3	.136	.137	.138	.139	.252
E1-2	.140	.141	.142	.143	.252
E3-2	.144	.145	.146	.147	.252
E3-4	.148	.149	.150	.151	.252
E4-5	.152	.153	.154	.155	.252
E5-2	.156	.157	.158	.159	.252

R1: 0/0 → PC4; 0/1 → R3; 1/0 → R2
 R2: 0/0 → PC2; 1/0 → R1; 0/1 → R5; 2/0 → R3
 R3: 0/0 → PC3; 0/1 → R1; 1/0 → R4; 2/0 → R2
 R4: 0/0 → PC5; 0/1 → R5; 1/0 → R3
 R5: 0/0 → PC1; 0/1 → R2; 1/0 → R4